

Unsupervised Anomaly Detection of Shipboard Electrical Power Consumption through GRU Autoencoder model

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Abstract— Anomaly detection (AD) in modern electrical microgrids is critical for ensuring reliable and safe operation as well as in improving the operational and maintenance efficiency of the infrastructure. This is particularly true for on-board microgrids, given their inherent electrical weakness and predominantly isolated operation. This paper proposes a machine learning-based approach for performing anomaly detection (AD) in electrical power consumption of a large passenger ship. The developed AD method relies on a gate recurrent unit (GRU) autoencoder (AE) model trained and validated with a multivariate electrical power time series collected from a real-world vessel. The proposed GRU AE approach is compared with different AE models, training and testing all models on an unsupervised context, and their performance is evaluated using various metrics suitable for such an unsupervised context.

Keywords—Anomaly Detection, Shipboard Electrical Power Consumption, Machine Learning, Autoencoder

I. INTRODUCTION

On-board Integrated Power Systems (IPS), currently used to provide both service and propulsion power in high-tech merchant and naval vessels, can be considered the same as an isolated microgrid; as such, they have low electrical inertia and are more susceptible and prone to failure than conventional power grids [1]. This limitation is likely to become even more severe, given the technological trend of increasing the shipboard integration of renewable energy sources to enhance the efficiency and environmental sustainability of shipping [2]. Ensuring reliable and safe operation, including proper identification of anomalous operating conditions, is a priority in such an application, particularly due to the need to safeguard passengers and crew[3].

Modern microgrids (regardless of whether for land or naval use) are composed of numerous smart devices. Such devices collect data, communicate with neighboring and remote devices, and can process information in order to increase the operational efficiency of the microgrid. This increased availability of data enables the analysis and identification of anomalous behaviors of the system.

Anomaly Detection (AD) is the problem of finding patterns in data that do not conform to expected behaviors [4]. Anomalies can occur in the data for a variety of reasons, such as malicious behavior, cyber-attack, malfunction (of the sensors or of the networking devices), system fault, non-

technical losses. Most anomalous behaviors in power grids can be detected by analyzing the data as time series, i.e., a series of data points indexed in time order.

Anomalies in time series data can be categorized as [5]:

- 1) Point Anomaly, characterized by a data point or a sequence that abruptly deviates from the trend.
- 2) Contextual Anomaly, where observations or sequences deviate from the expected patterns within the time series, however, if taken in isolation they may be within the range of values expected for that signal.
- 3) Collective Anomaly (also known as Pattern Anomaly), characterized by a collection of observations that are anomalous with respect to the rest of the data.

Typical examples of the above-described classes of anomalies are shown in Fig. 1 [6].

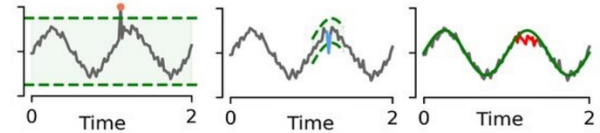


Fig. 1. Examples of point anomaly (left), contextual anomaly (middle), and collective anomaly (right) [6]

At this time few works have been published about AD in power systems onboard ships. These works, however, only focus on anomalies of the diesel engines. For example, in [7] a clustering algorithm was used to detect anomalies in sensor data from a marine diesel engine, whereas in [8] an ensemble-based approach is proposed to operate on a large-scale high-dimensional diesel engine sensor dataset.

The use of AD techniques developed for terrestrial power grids can be extended to onboard power systems. These techniques have been applied in smart grids and smart buildings for detecting, for example, abnormal end-user behavior, faulty appliances, premises occupancy, and non-technical losses [9]. It is worth observing that most of the works are focused on the consumption process and, more specifically, on the identification of anomalous behaviors in power load patterns [10].

Some early works applied mathematical and statistical models to AD in time series data. Generalized Extreme Studentized Deviation (ESD) was used in [11] to identify and remove anomalies within the energy consumption dataset of a commercial building. Furthermore, Autoregressive Moving

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Average (ARMA) was used in [12] to detect anomalous patterns in building office space energy consumption. However, these classical models lack the ability to deal with large and complex datasets and require expert domain knowledge to be fine-tuned. Machine learning and deep learning have been extensively used to address these limitations.

Machine learning-based AD methods (including deep learning methods) can be trained in supervised, semi-supervised, and unsupervised fashion. Supervised methods require labeled data in the training set to learn to detect anomalies and can also classify anomalies. Semi-supervised and unsupervised methods learn intrinsic patterns of the data and do not need labeled datasets for the learning process. Semi-supervised are trained exclusively on non-anomalous data, whereas unsupervised methods only assume that anomalies in the train dataset are rare.

Several different supervised methods have been applied to power consumption AD. A Support Vector Machine was used in [13] to detect false data injection in power flow measurements. Another approach combined micro-moments and a deep neural network classifier to detect anomalous power consumption in residential buildings [14]. Furthermore, many methods based on supervised Artificial Neural Networks (ANN) have been proposed. For example, Convolutional Neural Networks (CNN) were used in [15] and in [16] for energy theft detection. Also, the Long Short-Term Memory (LSTM) model was used in [17] to detect seasonal power consumption anomalies and in [18] for power consumption AD with the ability to distinguish anomalies from normal drifts in user behavior.

Semi-supervised and unsupervised learning methods have also found many applications in electrical power consumption AD, especially due to the freedom from the need of labeled datasets. It is assumed that anomalies are rare events, and the normal behavior is modeled from the training dataset. A clustering approach was proposed in [19] for AD of multi-user electricity consumption. Reconstruction-based methods have also been proposed in [20] where a principal component analysis-based method was used to detect anomalies in primary distribution voltage measurements. Furthermore, an AD method based on Autoencoders (AE) was proposed in [21] where reconstruction scores were used to identify anomalies.

Further interesting advances in time series AD are based on recent deep learning techniques. The attention mechanism and the Transformer architecture have been applied to AD in time series [22]. Furthermore, unsupervised generative models have found effective application in this domain as well. For example, Generative Adversarial Networks (GAN) were used for AD in the context of industrial machine data [23] and of treated water distribution systems [24]. Also, a diffusion model was used for AD and tested on time series AD benchmark datasets [25]. However, these methods still have not been applied to AD for power grids.

AD in power grids has seen many recent publications, but several challenges hinder its effective application. First of all, there is a need for widely accepted platforms to reproduce empirical results. Another challenge is due to the non-stationarity of real-world time series considering the constantly evolving power grids. Finally, there is a limited amount of data available. In particular, labeled data are

difficult to obtain and missing them for a test dataset makes binary classification metrics like precision and recall, very often used for AD, inapplicable, requiring the definition of surrogate metrics for unsupervised training and test.

In this paper the AD of shipboard electrical consumption is addressed for the first time. To this aim, an electrical power consumption dataset, consisting of electrical power measurements taken onboard a real-world ship over a period of four months, is used. The dataset does not provide anomaly labels.

The proposed method is based on a Gate Recurrent Unit (GRU) Autoencoder (AE) model which performs AD of a multivariate electrical power time series. The GRU AE model is also compared with different AE models, as for performance. Additionally, all the considered AE models are trained and tested on an unsupervised context, considering different surrogate metrics appropriate for such a context.

II. CASE STUDY

The case study considered for the investigation is the electric power consumption of the hotel and air conditioning loads of a large passenger ship with a total installed electric power of about 60,000 kW.

The shipboard power system is an IPS. A typical IPS outline single line diagram (SLD) is sketched in Fig. 2 [26]. The electrical power system on board involves a 11 kV AC primary distribution supplying air conditioning and, through a 690/220 V AC secondary distribution, several accommodation loads, galleys, and other auxiliary services.

The power consumption dataset, used to develop and validate the GRU AE-based AD technique, is a multivariate time series (containing 18 variables) which refers to electrical powers measured at each connector of the 11 kV main switchboards (AFT and FWD SWBDs), except for powers required by propulsion drives and bow thrusters. The aggregate power considered for this study is approximately the 25% of the total electrical power required by the ship; detecting anomalies in the time profile of this portion of the electrical load allows for an assessment of the ship's operating conditions particularly when it is moored in port.

The power time series data were collected with a 10 min. granularity in a four-month period, from September to December 2022.

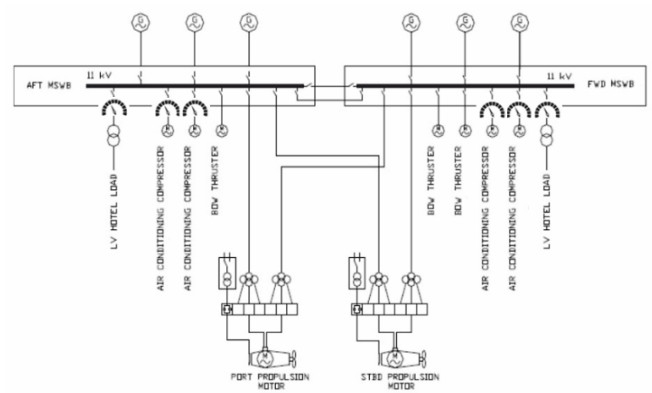


Fig. 2. Outline SLD of a shipboard IPS (the measurement points of power data used in this study are highlighted with schematic meters).

III. GRU AUTOENCODER MODEL FOR TIME SERIES AD

AEs are neural network models that learn latent representations of data and to reconstruct the input data from the latent representation [27]. Functionally, AE can be decomposed into an encoder that learns the function $z = f(x)$ and a decoder that learns the function $\hat{x} = g(z)$. These models are trained to minimize the reconstruction error $e = x - \hat{x}$ using regression losses like the mean squared error.

The considered dataset contains a multivariate time series with 18 variables or features. The primary assumption regarding the dataset, in an unsupervised AD fashion, is that anomalies are rare. As a result, our model can be trained to construct an appropriate latent space, thereby yielding a low reconstruction error (Fig. 3) when data are minimally affected by anomalies. Conversely, under conditions where data are substantially corrupted by anomalies, the model is expected to exhibit a high reconstruction error. A threshold can then be applied to identify time series subsequences as anomalous.

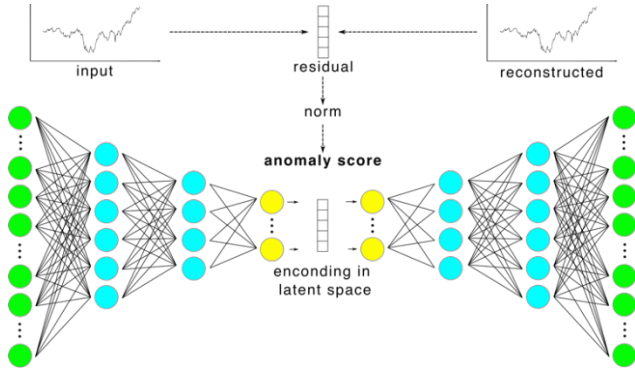


Fig. 3. Anomaly score calculation through reconstruction error of an AE model.

The proposed AD model for AD of electrical power consumption is based on the GRU RNN. This recurrent network uses gates that control whether to accumulate or forget certain information to learn long term relationships in sequence data. As shown in Fig. 4 the AD model uses a GRU layer as encoder, the latent representation is then presented as an input to another GRU layer and finally the reconstruction output is produced through a fully connected layer (FC) applied at each time step.

The main hyperparameters of the model are the number of recurrent units in the GRUs, which also define the size of the latent representation, and the length of the time series window input to the model. Values for these hyperparameters and the tuning procedure are described in Section V.

The threshold for anomaly detection can be manually selected or can be computed given the statistical distribution of the scores on the training set. The proposed approach is based on the latter and the threshold is computed as:

$$\tau = \mu + \sigma \quad (1)$$

where μ and σ are, respectively, the mean and standard deviation of the scores on the train dataset.

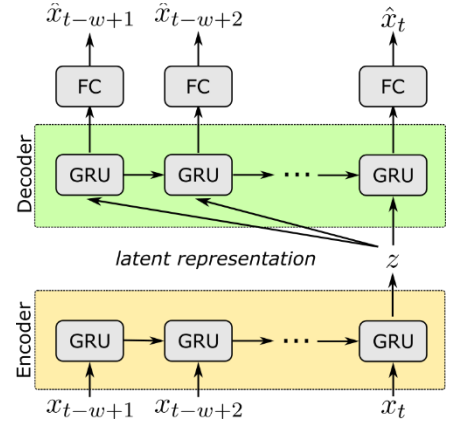


Fig. 4. GRU AE model architecture.

IV. UNSUPERVISED AD SURROGATE EVALUATION METRICS

Model selection for unsupervised AD is the problem of choosing the most accurate model among different models. However, in the absence of ground truth knowledge (i.e., labeled anomalies), it becomes essential to identify suitable surrogate metrics that correlate with model performance. Surrogate metrics proposed in [28], [29] do not consider the peculiarities of time series AD and are, therefore, inapplicable. Several surrogate metrics for time series AD are empirically evaluated and categorized in [30]. Considering that no single metrics can be considered universally superior to the others [31], the present work adopts the following two categories of metrics:

- **Reconstruction error metrics** – Metrics that consider the average reconstruction or prediction error across the entire test dataset. It is assumed that good reconstruction ability is a surrogate of good AD performance.
- **Synthetic anomaly injection metrics** – Several kinds of synthetic anomalies are injected into the test set and considering the knowledge of the injected anomalies supervised error metrics are applied.

Surrogate reconstruction error metrics applied in this work are the Mean Squared Error (MSE) and the Mean Absolute Error (MAE), both averaged over the test set. These are respectively defined as follows:

$$MSE = \frac{1}{T} \sum_{i=0}^{T-1} \|x_i - \hat{x}_i\|_2^2 \quad (2)$$

$$MAE = \frac{1}{T} \sum_{i=0}^{T-1} \|x_i - \hat{x}_i\|_1 \quad (3)$$

where vector $\hat{x}_i$ is the reconstructed value of vector x_i using the model under evaluation.

Surrogate metrics based on synthetic anomaly injection calculate the Area Under the Curve (AUC) of the Receiver Operating Characteristic (ROC), which is the curve representing the AD performance at varying thresholding values. It is the plot of true positive rate against false positive rate at each threshold value. The locations of the synthetic injected anomalies are used as labels to calculate true positives and false positives.

Three types of synthetic anomalies are injected: spike anomalies, noise anomalies, and cutoff anomalies. These are selected among the different types of anomalies discussed in [30], considering domain knowledge. Fig. 5 depicts these types of injected anomalies.

Spike anomalies are injected randomly choosing a n_{spikes} positions uniformly in the test dataset and then adding spikes with magnitude sampled from a gaussian with zero mean and σ_{spikes} standard deviation.

Noise anomalies are injected by randomly choosing up to n_{noise} random-length windows of the test set and adding gaussian noise with zero mean and σ_{noise} standard deviation to one of the variables of the time series.

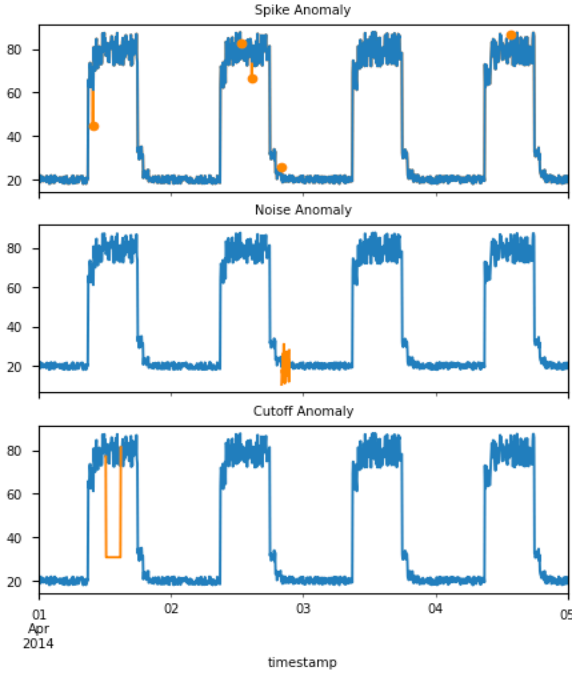


Fig. 5. Types of adopted synthetic anomalies. Visualized time series from dataset in [32].

Finally, cutoff anomalies are injected by randomly choosing up to n_{cutoff} random-length windows of the test set and setting all the values in the windows for a randomly chosen subset of the variables to a cutoff value. The cutoff value is sampled from a Rayleigh distribution with scale parameter σ_{cutoff} .

It is worth observing that surrogate metrics are applied directly to the anomaly scores produced on a test set. The threshold, and therefore the final binary detection result is not considered. In fact, for unsupervised model selection, threshold selection is considered an orthogonal problem [30], [31].

V. RESULTS

This section presents the performance of the proposed approach for AD of shipboard electrical consumption on the case study. The GRU AE model was trained and tested on the dataset along with other three AE models for comparison: a multilayer perceptron (MLP) AE, a convolutional AE, and an LSTM AE.

The dataset was split into train (the first two months), validation (the next month), and test sets (the last month). The

datasets were normalized so that every variable lies in the interval $[-1, 1]$ using the maximum and minimum values from the train dataset. Then, Fourier time features were added and windowed datasets were created with the procedure described in [33].

The AE models were trained for a maximum of 40 epochs using early stopping. The optimization algorithm used was Adam with exponential decay learning rate scheduler. The hyperparameters were in part manually tuned and in part optimized through grid search, a summary of the resulting hyperparameters is reported in Table I.

Training and tests were executed on a workstation with an Intel Xeon® w9-3475X processor, 128 GB of ECC RAM and an Nvidia RTX 6000 ADA GPU.

TABLE I. HYPERPARAMETERS OF THE MODELS USED IN THE EXPERIMENTS

Hyperparameter	Model Architecture			
	MLP AE	Conv AE	LSTM AE	GRU AE
Number of Fourier time features n	1			
Window size	12			
Initial learning rate	1.0e-3	8.5e-3	3.0e-4	8.5e-3
Dropout rate	0.0	0.2	0.0	0.0
Recurrent units	-	-	256	128
FC units	32	-	-	-
Num filters	-	32	-	-
Kernel size	-	3	-	-
Pool size	-	2	-	-

The resulting values of the adopted surrogate metrics for each model architecture on the test set are reported in Table II. It can be observed that RNN-based AEs perform better on almost all the metrics. Of these, GRU AE shows the best performance. This result can be attributed to the simplified gating architecture that allows better learning on such a small dataset.

It can be observed that none of the AUC-ROC values calculated for synthetic anomaly injection types, which can vary between zero and one, exhibit remarkable performance when viewed in isolation. For synthetic noise anomalies, all values fall below 0.5. However, these surrogate metrics are intended for model selection and comparison. Thus their values should not be considered in isolation to determine model performance.

Hyperparameter tuning revealed that for all the considered AE architectures, except that for the convolutional AE, using dropout led to worse models, according to the adopted surrogate metrics. This observation is counter intuitive given that dropout typically exerts a regularizing effect that should improve training in the presence of outliers (i.e., anomalies in this context). Consequently, the impact of dropout on surrogate metrics for AEs warrants further investigation.

TABLE II. PERFORMANCE OF THE PROPOSED MODEL AND THE OTHER MODELS WITH RESPECT TO THE ADOPTED SURROGATE METRICS.

Surrogate metric	Model Architecture			
	MLP AE	Conv AE	LSTM AE	GRU AE
MSE ^a	14.1586	15.2662	13.1872	12.5283
MAE ^a	0.7845	0.7825	0.6239	0.3842

Surrogate metric	Model Architecture			
	MLP AE	Conv AE	LSTM AE	GRU AE
Synth. Spikes AUC-ROC ^b	0.4235	0.4674	0.3901	0.4238
Synth. Noise AUC-ROC ^b	0.3352	0.4812	0.2352	0.2171
Synth. Cutoff AUC-ROC ^b	0.3855	0.2313	0.5095	0.5871

^a. Lower values are better; ^b. Higher values are better

Finally, to provide a graphical example of the results, Fig. 6 shows anomalies identified by the GRU AE on a subset of the test set. Only a sample of the variables are depicted, i.e., powers consumed by two engine room (ER) service loads, one accommodation zone, and one galley. Data are normalized according to train dataset minimum and maximum values as described in [33]. The threshold was selected according to (1). It can be observed that one of the variables (ER 2) is clearly outside the normal bounds. The electrical load associated with this variable is almost inactive except during the time windows in which the anomalies were detected.

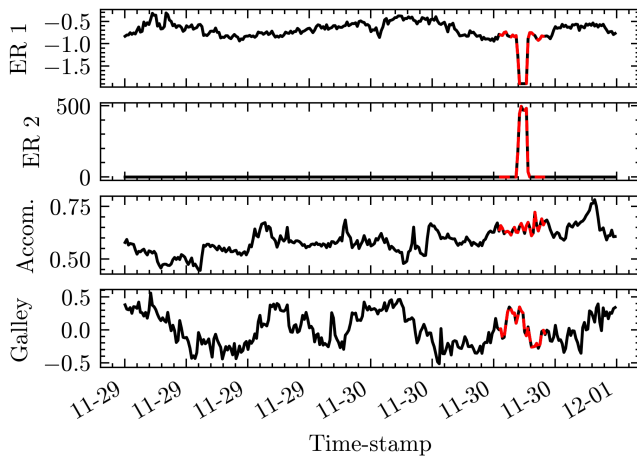


Fig. 6. Identified anomalous windows on a two-day subset of the test dataset.

VI. CONCLUSIONS AND FUTURE WORK

This paper proposes a machine learning-based approach developed to perform AD of electrical power consumption on board a large passenger ship. This work fills a gap in the electrical literature since contributions on electrical load AD published so far refer only to the case of land-based electrical power grids. The main contributions of this work can be summarized as follows:

- A GRU AE model was developed for AD of a multivariate time series of a real-world ship electrical power consumption.
- Performance of the proposed model was assessed and compared with that of several other AE models.
- All the AE models were trained and tested on an unsupervised context and evaluated according to different surrogate metrics appropriate to such an unsupervised context.

The proposed GRU AE model demonstrated its ability to detect anomalies in the examined shipboard electricity consumption time series, outperforming the other AE models in terms of all the adopted surrogate metrics, except for synthetic noise anomaly injection metric.

Further research includes the evaluation of other AE model architectures and the addition of a model ranking approach able to combine the different surrogate metrics.

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